

Python Pandas Interview Questions



Given the `df_users_interactions` and `df_user_profiles`, solve the problem set using the Pandas library. You may use Numpy, if necessary, but no other libraries.

`df_user_interactions.csv`

user_id	date	pages	time_spent	platform
1001	2023-01-05	3	15	Mobile
1002	2023-01-06	5	20	Web
1003	2023-01-07	4	18	VR
1004	2023-01-08	6	25	Web

`df_user_profiles.csv`

user_id	age	gender	country	registered_since
1001	28	M	US	2020-06-01
1002	34	F	UK	2018-11-15
1003	24	M	IN	2021-05-20

1. Using the ``df_user_interactions`` and ``df_user_profiles`` DataFrames, determine the country of each user and their average time spent on interactions. Only include users in both DataFrames.
2. Given the User Profiles (``df_user_profiles``) DataFrame, find out how many users have registered in each year. Display the counts in descending order.

∞ Solutions #1



1. Using the `df\_user\_interactions` and `df\_user\_profiles` DataFrames, determine the country of each user and their average time spent on interactions. Only include users in both DataFrames.

```
df_user = (  
    # Start with the interactions data  
    df_user_interactions  
    # Merge with user profiles based on user_id  
    .merge(df_user_profiles, how='inner', on='user_id')  
    # Group by user_id and country  
    .groupby(['user_id', 'country'])['time_spent_minutes']  
    # Calculate the average time spent  
    .mean()  
    # Reset the index for a clean DataFrame format  
    .reset_index()  
)
```

Output Table

```
+-----+-----+-----+  
| user_id | country | avg_time_spent_min |  
+-----+-----+-----+  
| 1001    | US      | 15                  |  
| 1002    | UK      | 20                  |  
| 1003    | IN      | 18                  |  
+-----+-----+-----+
```

∞ Solutions #2



2. Given the User Profiles (`df\_user\_profiles`) DataFrame, find out how many users have registered in each year. Display the counts in descending order.

```
# Convert the 'registered_since' column to datetime format
df_user_profiles['registered_since'] = (
    pd.to_datetime(df_user_profiles['registered_since'])
)

# Extract the year from the 'registered_since' column and count
# the occurrences
yearly_counts = (
    df_user_profiles['registered_since']
    .dt.year.value_counts()
)

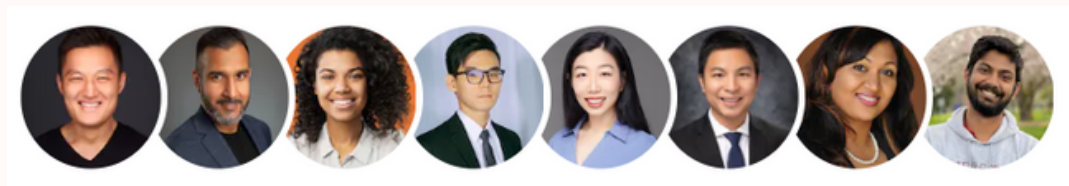
# Sort the counts in descending order
sorted_counts = yearly_counts.sort_values(ascending=False)
```

Output Table

year	user_count
2021	1
2020	1
2019	1
2018	1

Join DataInterview.com

Access courses, mock interviewing and private community with coaches and candidates like you



**Created by interviewers at top companies
like Google and Meta**

